UTM UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SCSM2113 - HUMAN-COMPUTER INTERACTION

SEMESTER III, SESSION 2025/2026

Project Part 3: Conceptual and Physical Design

Lecturer: Prof. Madya Ts. Dr. Masitah binti Ghazali

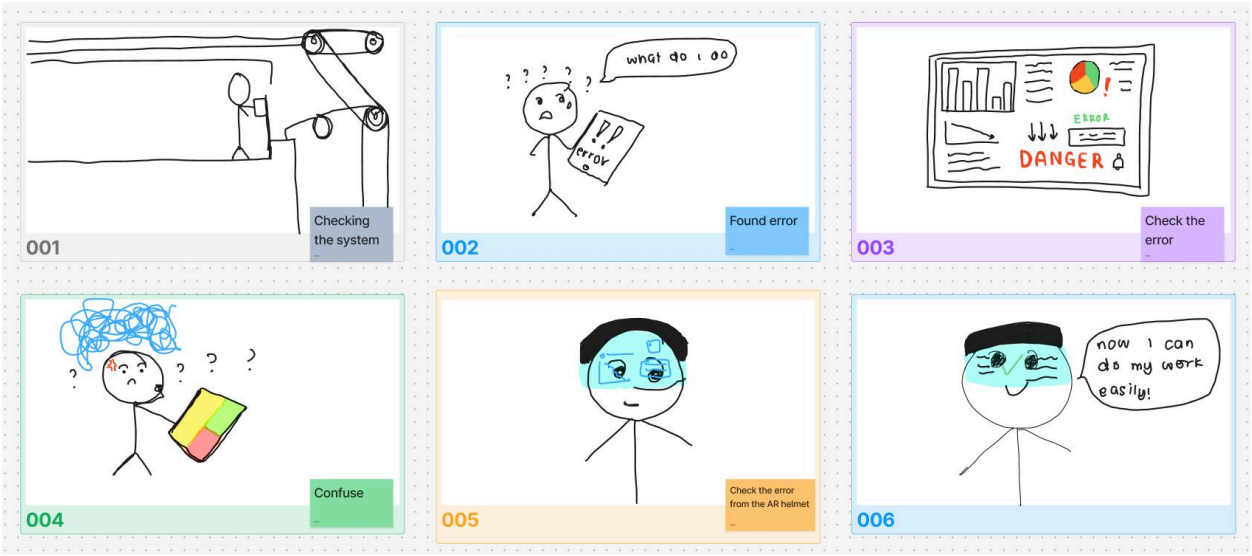
Section: 01

Group: Team 3

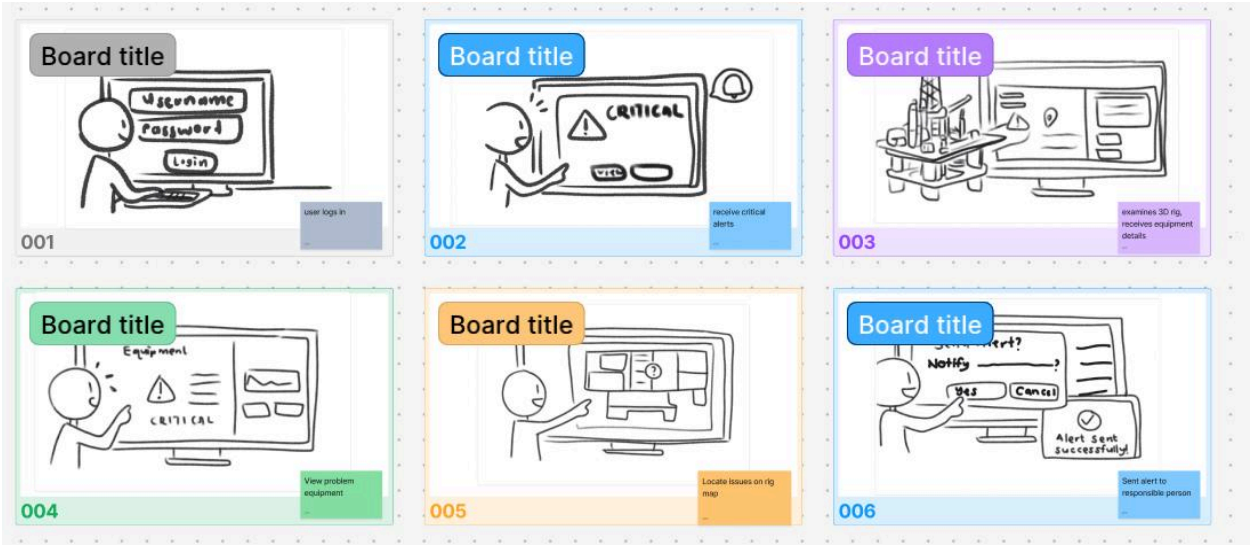
NAME	MATRIC NO
ALVANNUR KHOIR TSAURI	A24CS0006
MOHAMMED MOHSEN MOHAMMED AL-SAKKAF	A23CS4026
KUGAANN SURESH	A24CS0098
AININ SOFIYA BINTI ZUKHRI	A24CS0040

STORYBOARDS

1. **Maintenance Engineer**
   Task 1: Monitor Equipment or System Information

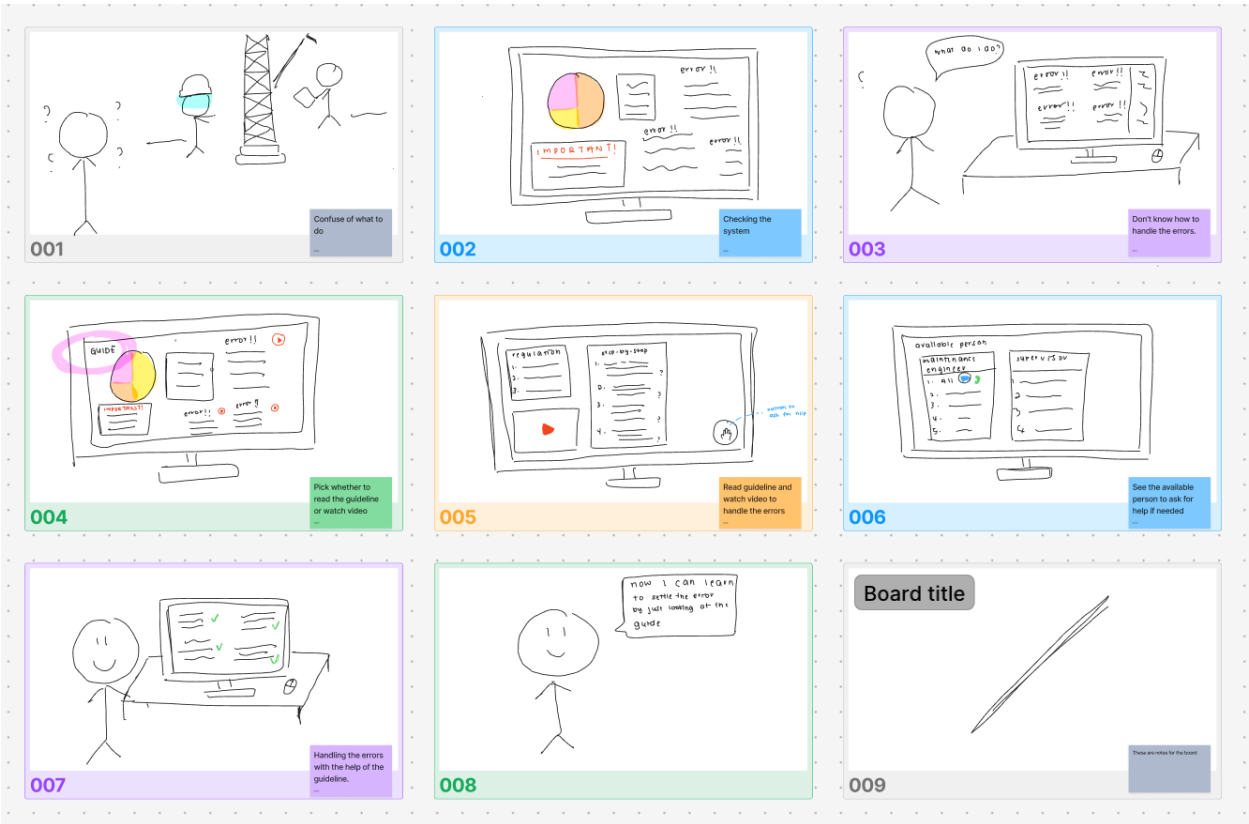


2. **Supervisor**
   Task 2 : Locate Issues on Equipment or Around the Rig



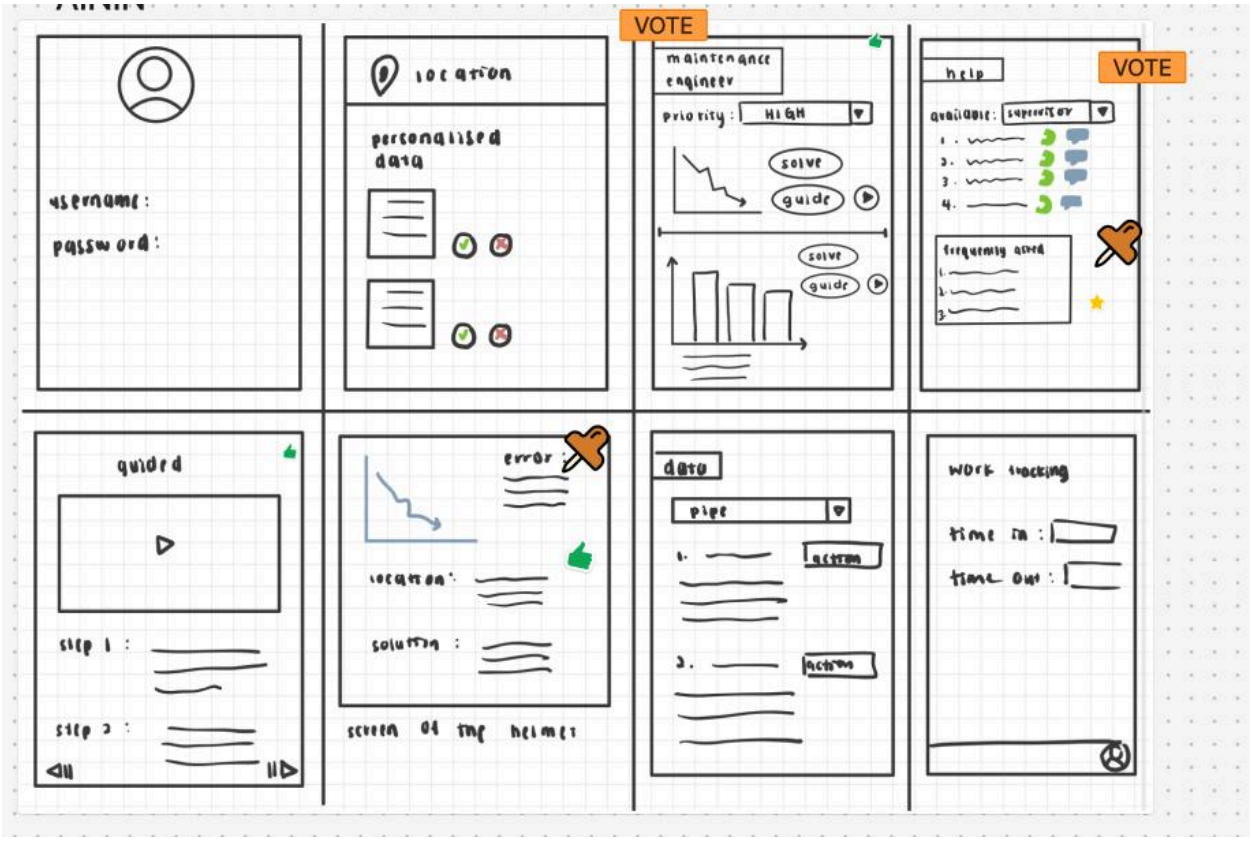
3. **Trainee**

Task 3 : Follow Structured Steps and Procedures to Perform Task Safely

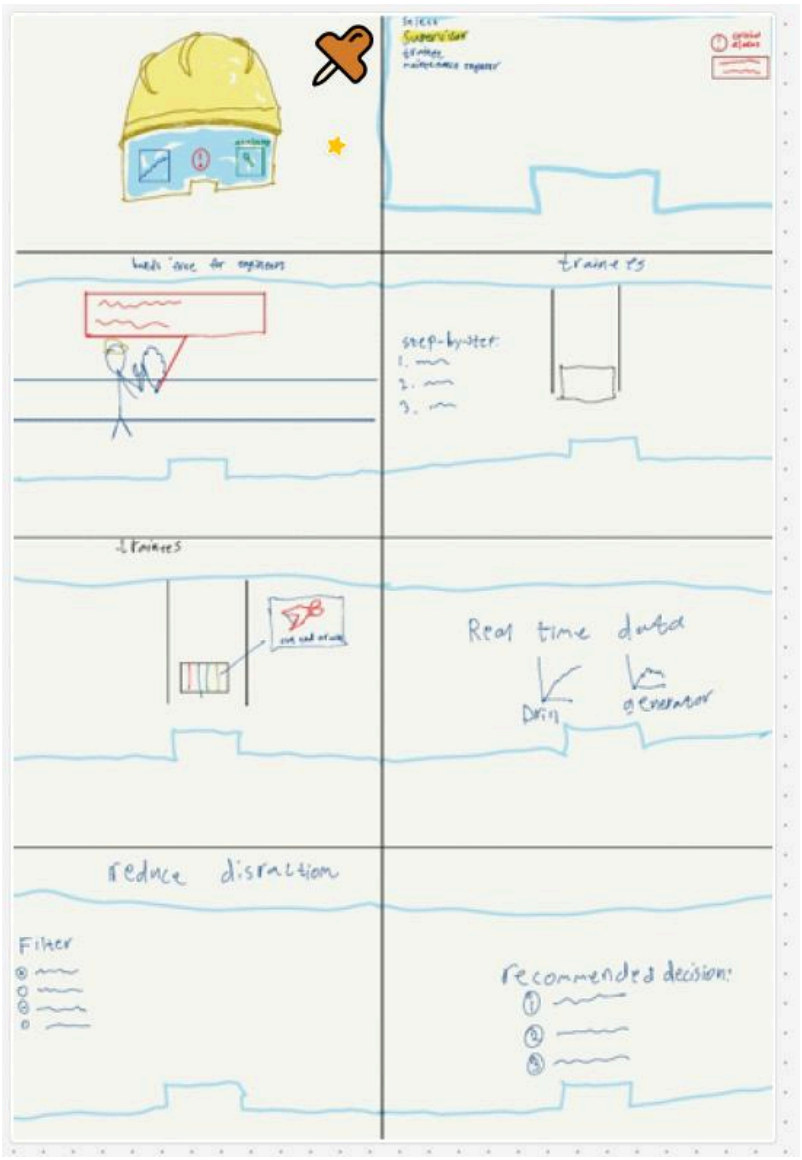


ALTERNATIVE DESIGN: CRAZY 8

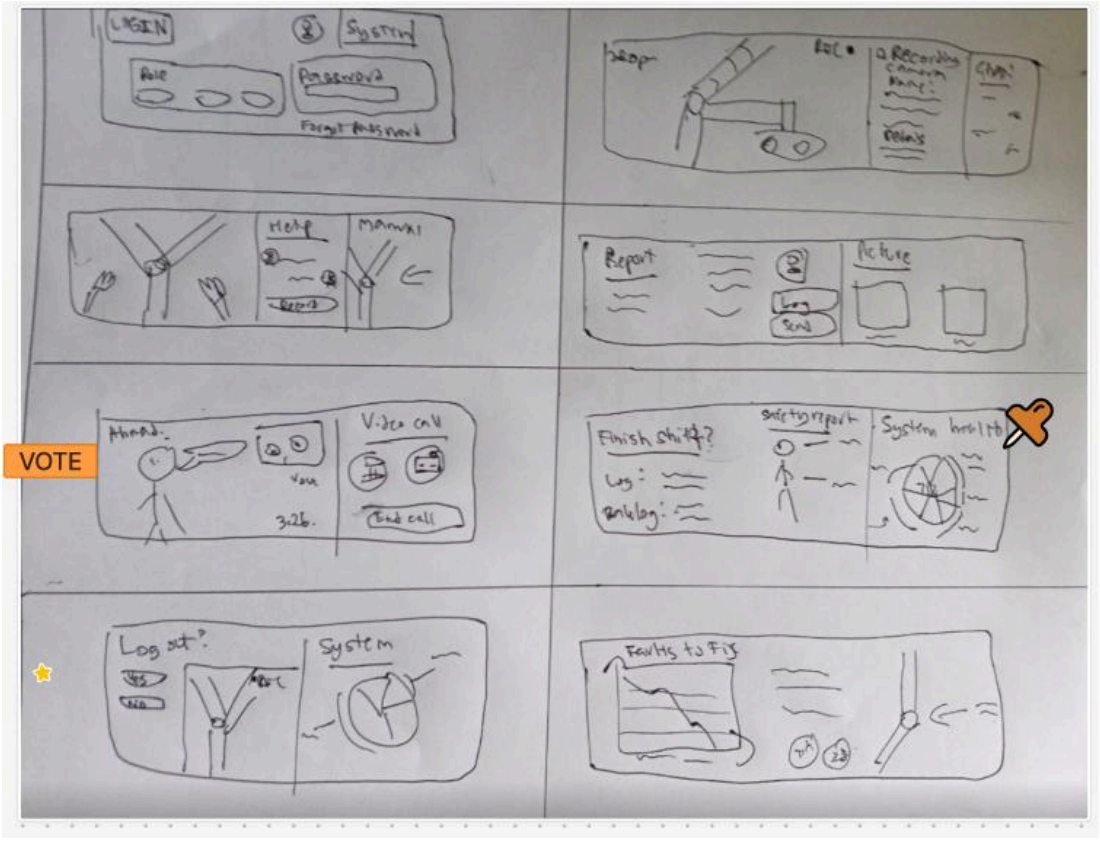
1) **Ainin**



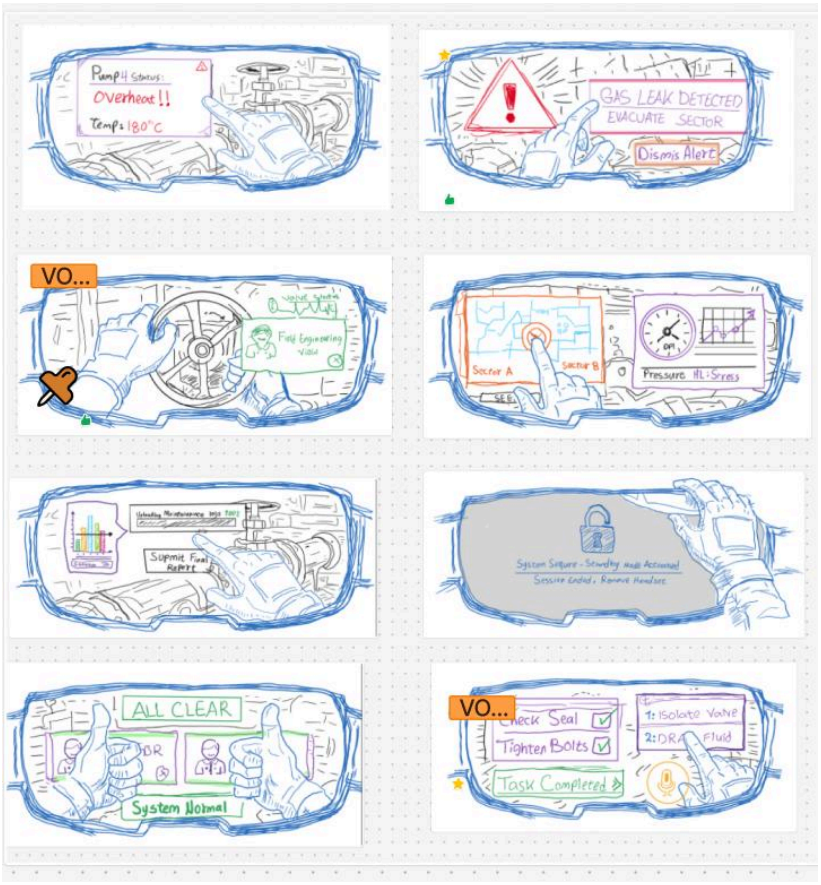
2) **Alvannur**


trainers
step-by-step:
1.
2.
3.
Trainees
Real time data
Data
Operator
reduce distraction
Filter
recommended decision:
1
2
3

3) Kugaann

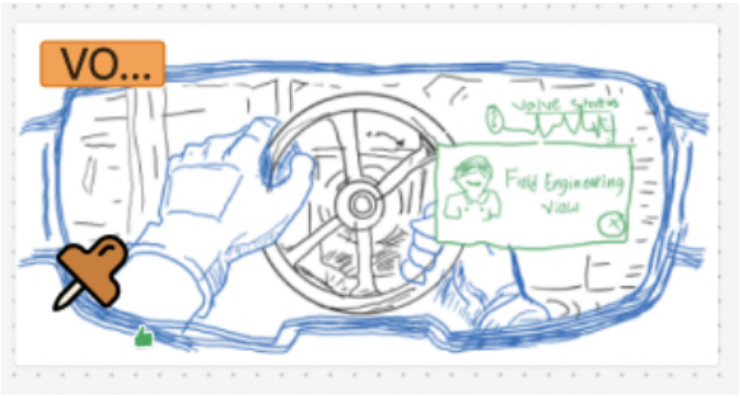

LOGIN
System
Help
Manual
Report
Video call
3:26
End call
Finish shift?
Log:
Debrief:
Safety report
System health
Log out?
System
Faults to Fix
VOTE

4) Alsakkaf



For the Crazy 8 method, all of the members are required to vote for 5 interfaces that they find might have effective features for our Digital Twin Project of Oil Rig Sector. Each of us picks a different type of sticker and puts it on the interfaces that we are interested in.

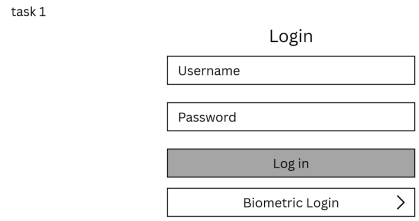
Based on the voting, the highest number of picks with 4 votes each goes to :


VO...
Field Engineering

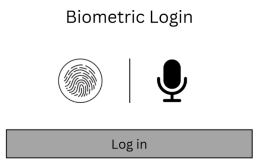

help
VOTE
frequently asked

WIREFRAMES

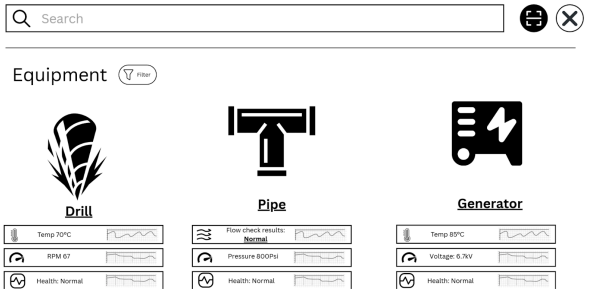
Task 1: Monitor Equipment or System Information



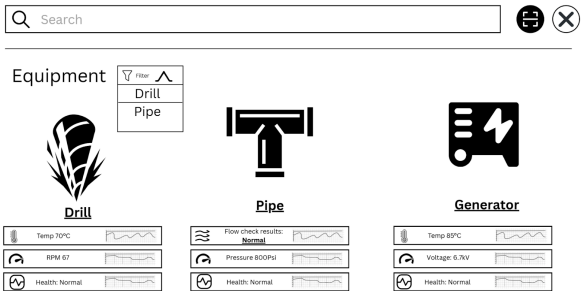
Employee login by inserting their username or password.



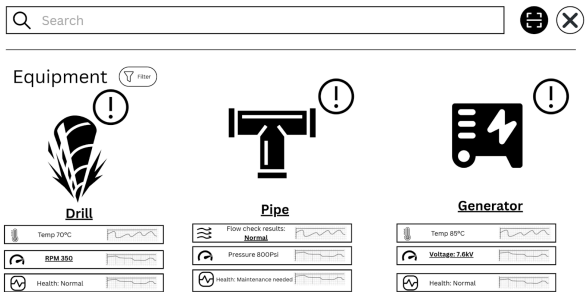
If the employee selects the Biometric Login button they are taken to a page where they login with either voice or fingerprint.



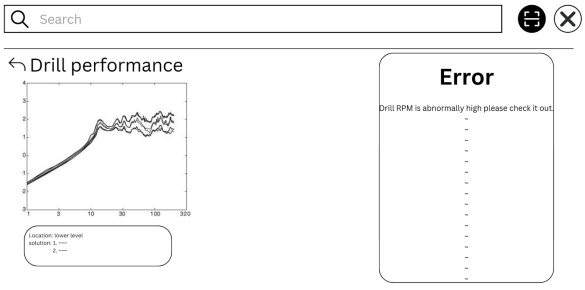
Then they are taken to the equipment page in which they can monitor different equipments and see its status.



When they select the filter button the employee can choose what equipment they want to specifically monitor.

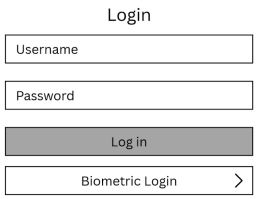


When there is abnormality to the equipment the system will notify the user and highlight which equipment(s) is having problems.

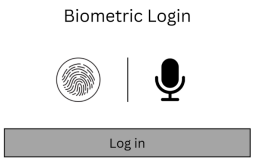


If they select on the abnormal equipment they are able to see the error description, suggested solution, and also the location of the equipment.

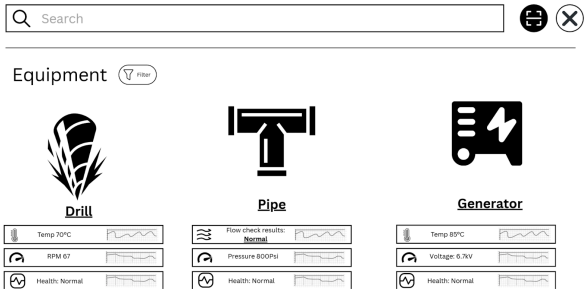
task 1



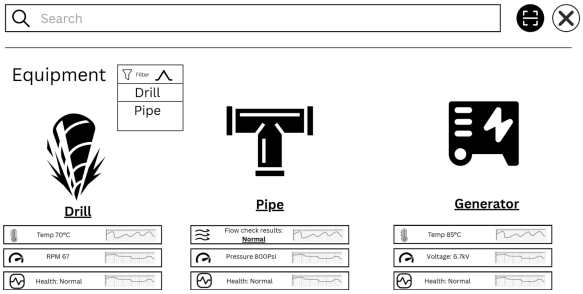
Employee login by inserting their username or password.



If the employee selects the Biometric Login button they are taken to a page where they login with either voice or fingerprint.



Then they are taken to the equipment page in which they can monitor different equipments and see its status.



When they select the filter button the employee can choose what equipment they want to specifically monitor.



And for maintenance engineers that want to do maintenance on the equipment they can select the scan button to see what's wrong and what recommended steps they can take.

Task 2 : Locate Issues on Equipment or Around the Rig



Step 1: The Supervisor logs in by entering their username and password to access the secure system

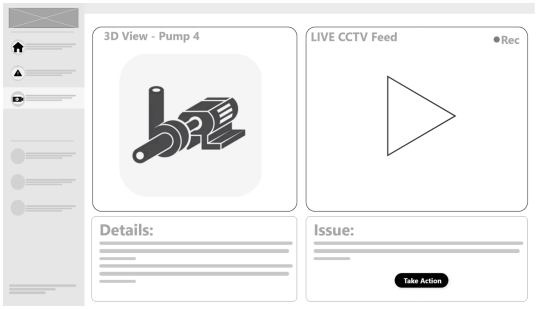
Step 2: If the supervisor selects the Biometric Login button, they are taken to a page where they verify their identity using facial recognition or fingerprint for quick access

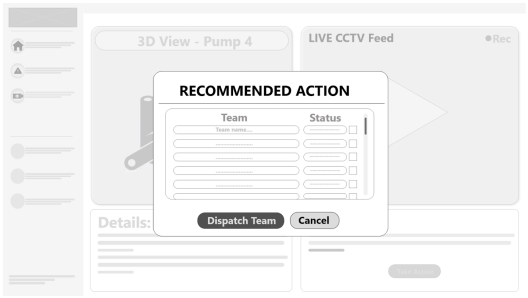




Step 3: Then they are taken to the main Dashboard where they monitor system health. A Critical Alert pops up at the top to warn them about specific equipment issues such as Pump 04 overheating

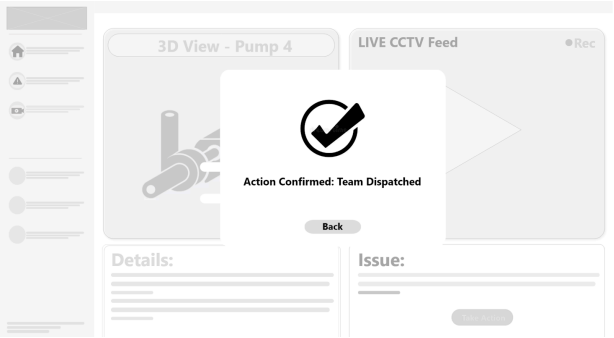
Step 4: When the supervisor selects the critical alert. They are directed to the Zone Summary page. The system maps out the rig's layout and highlights Zone A as (Critical) allowing the supervisor to pinpoint the exact location of the faulty Pump 04 immediately



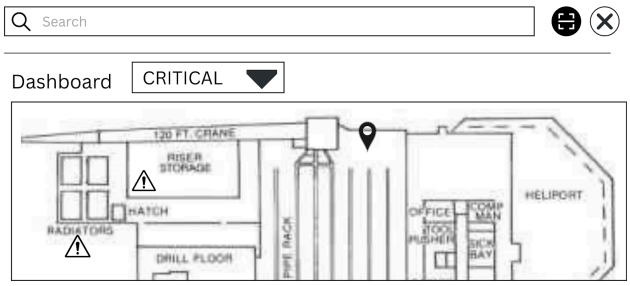


Step 5: The screen displays a split view: a 3D Model of the pump and a Live CCTV Feed for visual verification so after reviewing the specific fault details the supervisor clicks the Take Action button to proceed with assigning a maintenance team to fix the issue

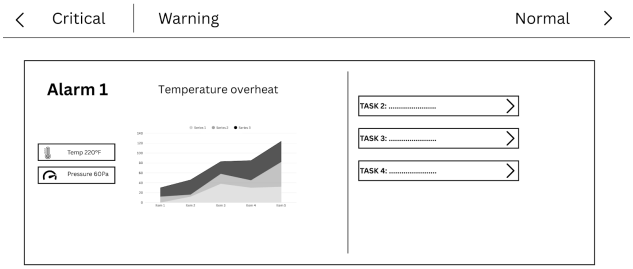
Step 6: Then a Recommended Action pop-up appears overlaying the screen displaying a list of maintenance teams and their availability status. The supervisor selects the appropriate team and clicks (Dispatch Team) to confirm the assignment



Step 7: Finally, the system shows a message saying (Action Confirmed: Team Dispatched)
This tells the Supervisor that help is on the way. Once she sees this, she clicks the (Back) button to return to the main monitoring screen



Step 8: After clicking (Back) the Supervisor returns to the main Facility Map. Here, she views the whole building plan to check for other problems in different areas like the Drill Floor. She can use the menu to filter and see only the (Critical) zones



Step 9: If she clicks on a specific alert from the map, the system opens the Alarm Details page. This screen shows simple graphs for temperature and pressure to analyze the problem On the right it lists the specific Tasks she must follow to finish the job completely

Task 3 : Follow Structured Steps and Procedures to Perform Task Safely



Step 1: The Supervisor logs in by entering their username and password to access the secure system

Step 2: If the supervisor selects the Biometric Login button, they are taken to a page where they verify their identity using facial recognition or fingerprint for quick access

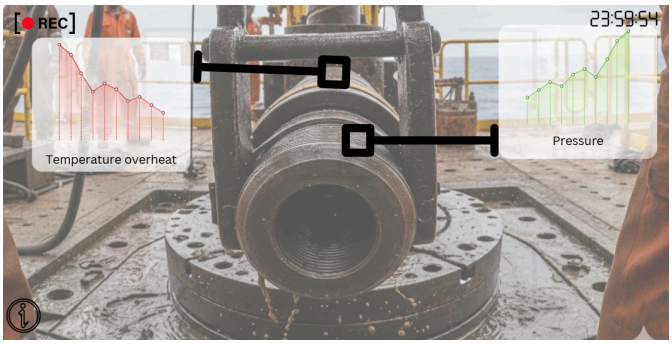




Step 3: The Supervisor is able to view the helmet which broadcasts the perspective of the oil rig workers and has the option to call the workers if they need assistance in conducting the task

Step 4: The Supervisor is able to record the session and call the oil rig worker via video conferencing and waits for the oil rig worker to accept.



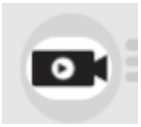
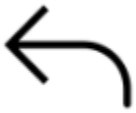


Step 5: The Supervisor is able to communicate with the oil rig worker on how to execute the task safely.

Step 6: The Supervisor is identify the statistics of the machine that is being viewed by the oil rig worker at the moment through the “i” symbol.

INTERACTION METAPHORS

	**Fingerprint** **Function:** for biometric login or identification
	**Microphone** **Function:** for voice input or commands
	**Magnifying glass** **Function:** for searching something
	**Scanner** **Function:** used for scanning the image of the equipment
	**Cross signed** **Function:** symbolizes an error or close action
	**Exclamation mark** **Function:** indicates an alert or warning
	**Dangerous signed** **Function:** used for critical alerts or danger notification

	**Arrow** **Function:** use to continue to the next part
	**Home** **Function:** used for returning to the main screen
	**Recorder** **Function:** represent a tool for recording
	**Back** **Function:** use to go to the last page.
	**Filter** **Function:** use to narrow down/focus on specific equipment.
	**Info** **Function:** press to obtain statistics on the equipment handled by oil rig